

# ARYAN RANE

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## Professional Summary

Information Technology student with hands-on experience in Full Stack Development, Cloud Computing, Data Analytics, and Machine Learning. Skilled in React.js, Node.js, Python, AWS, Docker, MongoDB, and Power BI; experienced building scalable web apps, cloud-native systems, and data-driven solutions.

## Education

**Shah & Anchor Kutchhi Engineering College**

BTech – Information Technology

**GPA: 8.0+ / 10.0**

Mumbai, India

*Currently in 3rd Year*

## Technical Skills

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|-----------------------------|----------------------------------------------------------------------------------|
| <b>Languages</b>            | JavaScript, TypeScript, Python, C++, C                                           |
| <b>Frontend</b>             | React.js, React Router, Tailwind CSS, Framer Motion, D3.js, Recharts, Chart.js   |
| <b>Backend</b>              | Node.js, Express.js, Flask, MongoDB, MySQL, JDBC                                 |
| <b>Cloud &amp; DevOps</b>   | AWS (Lambda, S3, API Gateway), Docker, Jenkins, Prometheus, Grafana, Git, Vercel |
| <b>ML &amp; NLP</b>         | scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Streamlit, TF-IDF, Naive Bayes |
| <b>Data &amp; Analytics</b> | Power BI (Advanced), Plotly                                                      |
| <b>Security</b>             | AES-256 Encryption, Cryptography Fundamentals, Computer Networks                 |

## Projects

**Legal Rights Awareness Portal** *React 18, Node.js, Express.js, MongoDB, Docker, Jenkins, Selenium, Prometheus, Grafana*  
— *Live on Vercel*

- Built a full-stack legal education portal covering cybercrime, data protection, and student rights, deployed live on Vercel with a complete CI/CD DevOps pipeline.
- Implemented Docker containerisation, Jenkins CI/CD, Selenium automated testing, and Prometheus + Grafana monitoring; managed sprints via Jira.
- Features interactive flashcard games, scenario-based legal quizzes, and JWT-authenticated user sessions; aligned with SDG 16 (Justice & Strong Institutions).

**AegisNet AI – Enterprise Network Intrusion Detection Dashboard** *React 18, TypeScript, Vite, Tailwind CSS, D3.js, Recharts, Framer Motion, Supabase, Chrome MV3*

- Built a real-time AI-powered intrusion detection dashboard using an Isolation Forest ML model to score every network packet, with XAI forensics for per-packet feature importance.
- Implemented NLP-based threat classification using TF-IDF and Naive Bayes; built an intent-aware SOC chatbot with semantic search – entirely browser-native, no external API calls.
- Includes a Chrome Extension (MV3) for live browser traffic capture, MITRE ATT&CK mapping, CVE references, attack replay for 7 attack types, and one-click firewall mitigation.

**CloudFrame – Serverless Image Upload & Processing System** *AWS Lambda, Amazon S3, API Gateway, Node.js, IAM, HTML5, CSS3, JavaScript*

- Designed and deployed a fully serverless, event-driven image processing pipeline on AWS – uploads via pre-signed S3 PUT URLs trigger Lambda functions that auto-resize images and generate WebP thumbnails.
- Achieved 100% test case pass rate across all 8 scenarios; warm Lambda invocations averaged 2.1s. Enforced least-privilege IAM, 5-minute pre-signed URL TTL, and 7-day S3 lifecycle expiry.

**SentiScope – Twitter (X) Sentiment Analysis** *Python, Flask, scikit-learn, TF-IDF, Logistic Regression, joblib, Chart.js, Pandas*

- Trained a scikit-learn Logistic Regression + TF-IDF pipeline on 50,000 tweets (Sentiment140) with bigram feature extraction, L2 regularisation, and model serialisation via joblib.
- Deployed as a Flask web app returning real-time sentiment predictions with per-class confidence scores visualised via Chart.js; custom tweet-cleaning pipeline handles URLs, mentions, and hashtags.

## **Air Quality & Weather Monitoring and Forecasting** *Python, Streamlit, Pandas, NumPy, scikit-learn, Plotly, Meteostat API*

- Built a Streamlit dashboard merging synthetic air quality (PM2.5, PM10, NO2) and real Mumbai weather data, performing EDA with correlation analysis between wind speed, humidity, and pollutant levels.
- Trained Linear Regression and Random Forest models for pollutant forecasting; predicted graphs closely matched actual pollution trends.

## **FlavourMap – Indian Food Dataset EDA & Visual Analytics** *Python, Pandas, Matplotlib, Seaborn, Power BI, Google Colab*

- Performed full EDA on 6,871 Indian recipes (Kaggle) – including custom time-string parsing, null removal, and column standardisation – generating 6 Python charts and an interactive Power BI dashboard.

## **Certifications & Courses**

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- **AWS Academy Graduate – Cloud Foundations** | Amazon Web Services · April 2026 · 20 hours
- **Python Essentials 1** | Cisco Networking Academy / OpenEDG Python Institute · April 2025
- **Networking Basics** | Cisco Networking Academy · December 2024
- **Introduction to Cybersecurity** | Cisco Networking Academy · July 2024
- **Programming Essentials in C (CLA)** | Cisco Networking Academy · April 2024
- **Game Development Using AI** | CogniScience Club, CSI-SAKEC & SAKEC-ACM · March 2025
- **Design Thinking Workshop** | Entrepreneurship & Innovation Cell, SAKEC

## **Achievements & Activities**

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- Fullstack Fusion Web Development Hackathon Participant (2024) – CogniScience Club & SAKEC AI&DS
- Research paper authored on Digital Legal Rights Awareness Portal – submitted and under revision for resubmission
- AWS Academy Graduate – Cloud Foundations (April 2026)
- Proficient in full-stack development, cloud computing, and data analytics across 10+ academic and personal projects